



## Full length article

# What kind of news gatekeepers do we want machines to be? Filter bubbles, fragmentation, and the normative dimensions of algorithmic recommendations

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## ABSTRACT

Machines are increasingly aiding or replacing humans in journalistic work, primarily in news distribution. We examined whether news recommendation engines contribute to filter bubbles and fragmented news audiences by asking a diverse set of real-world participants ( $N = 168$ ), using their personal Google accounts, to search Google News for news about Hillary Clinton and Donald Trump during the 2016 U.S. presidential campaign and report the first five stories they were recommended on each candidate. Users with different political leanings from different states were recommended very similar news, challenging the assumption that algorithms necessarily encourage echo chambers. Yet we also found a very high degree of homogeneity and concentration in the news recommendations. On average, the most recommended five news organizations comprised 69% of all recommendations. Five news organizations alone accounted for 49% of the total number of recommendations collected. Out of 14 organizations that dominated recommendations across the different searches, only three were born-digital, indicating that the news agenda constructed on Google News replicates traditional industry structures more than disrupts them. We use these findings to explore the challenges of studying machine behavior in news from a normative perspective, given the lack of agreed-upon normative standards for humans as news gatekeepers. This article suggests that because there is no one agreed-upon standard for humans as news gatekeepers, assessing the performance of machines in that role is doubly complicated.

## 1. Introduction

In much of academic and popular discourse, there is a present fascination with machines—in particular, machine processes and products that not only facilitate human activity, as personal computers have been doing for decades, but also augment and extend it, even placing machines in roles previously associated with humans alone (Brynjolfsson & McAfee, 2016). Such machines are part of a broader turn toward “smart” technologies and artificial intelligence in developed societies (Howard, 2015). They increasingly stand in for humans across many sectors, from Siri and Alexa personal assistants to customer service chatbots, often raising concerns about the “threat of a jobless future” amid the rise of robot replacements (Ford, 2016). Whether to be feared or embraced, these developments are particularly visible in information work such as journalism, where machines—in the form of algorithms, applications, and automated systems—play a growing role in determining what kind of news is produced, where it appears online,

and how it reaches some audiences and not others. In effect, machines are taking on more of the news gatekeeping functions—of story selection, story placement, and so on—that were formerly controlled by human editors (Shoemaker, Vos, & Reese, 2008).

Algorithms are a definitive element of today's media technologies (Gillespie, 2014), and “algorithmic gatekeepers” (Napoli, 2015) have been a key feature of digital journalism at least since the 2002 introduction of Google News, which was the first to organize headlines without human intervention. Such systems and processes have come to define social media platforms such as Facebook that prioritize content, including the visibility of news content, based on user profiles and preferences. By now, there is little debate about the efficiency and consistency of machine gatekeepers relative to human counterparts: they can do more, at scale, reliably. Nevertheless, while there is broad recognition of what machines can do, there is growing unease about what they *should* be doing, both generally in how algorithms increasingly shape social decisions about jobs, loans, welfare assistance, and

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prison sentences (Eubanks, 2018; O’Neil, 2016) and in how they adjudicate matters of news and information in public life (Diakopoulos, 2015; Lewis, Sanders, & Carmody, 2018).

In communication and media studies, emerging research examines not only what machines are doing as news gatekeepers, but also why they are being deployed in particular ways—and how they might better serve longstanding goals of civil discourse, democratic values, and the public interest (see, e.g., discussion in Gillespie, 2014; Napoli, 2015). In short, the unique role of news in society, as a facilitator of knowledge about public affairs (Schudson, 1995), leads to a fundamental normative question: Exactly what kind of gatekeepers do we want machines to be?

This concern comes at a time when algorithms appear more prominent than editors in selecting the news people see online. More than half of news users globally (54%) prefer algorithmic forms of news presentation (i.e., search engines, social media, and aggregators) as compared to those handled by journalists (i.e., via news site/app, email, and mobile notifications), and that preference is stronger for those who mainly use smartphones (58%) and are younger (64%) (Newman, Fletcher, Kalogeropoulos, Levy, & Nielsen, 2017). Through a combination of deliberate and incidental exposure to news online, Google and Facebook together have been estimated to drive roughly 60% of the traffic to news sites in recent years (Tran, 2018). News aggregators in particular have grown substantially in recent years, with Google News—which blends forms of Google search with the companion aggregation site news.google.com—standing apart as the most popular form of algorithm-as-editor (Newman et al., 2017). With an estimated 500 million visits per month, Google News searches have been found to provide large numbers of referrals to prominent news sites—in some cases accounting for more than 30% of a site’s traffic (Sherk, 2014).

As such, this study addresses the normative tension around machines-as-gatekeepers in two ways: First, we develop and implement an innovative method for measuring personalization in algorithmic news recommendations. Using a real-world setting, we ask a diverse set of participants ( $N = 168$ ) to search Google News for news about Hillary Clinton and Donald Trump during the 2016 election campaign. Second, we build upon the findings of that empirical study to illuminate the particular challenges facing multiple stakeholders—algorithm designers, journalists, media scholars, and users alike—in determining what counts as “good” conduct by machines acting as news gatekeepers. Because journalism is already such contested terrain, with no one set standard for human behavior, establishing the normative priorities for machines is especially fraught. In closing, we use this two-part examination to develop a research agenda for the study of machines and their normative role in the public information environment.

## 2. Gatekeeping and the ‘presentational authority of journalism’

The study of journalism has long been a study of the forces—ideological, institutional, organizational, political-economic, and so forth—that shape what counts as news, who gets to make it, and how it is arranged for public consumption (see, e.g., Gans, 1979; Reese & Shoemaker, 2016; Tuchman, 1978). Much of this media sociology research, from early studies of social control in the newsroom (Brede, 1955) to contemporary accounts of newspapers transitioning online (Ryfe, 2013), has focused on the people behind these processes, examining how and why journalists, as gatekeepers, make certain decisions about what to cover, what to prioritize, and why. Indeed, from its earliest studies (White, 1950), gatekeeping was conceptualized as a distinctly human activity, one focused on “the process of selecting, writing, editing, positioning, scheduling, repeating, and otherwise massaging information to become news” (Shoemaker et al., 2008, p. 73). Prominent among these functions is what Carlson (2007, p. 1015) calls the “presentational authority of journalism,” or the ability to generate cultural meaning through the purposeful ordering, arranging, and highlighting of some news items relative to others. Human editors,

by determining factors such as story length, headline size, and placement on a newspaper page or in a broadcast, could amplify certain stories over others. As Schudson (1995) describes:

A news story is an announcement of special interest and importance. It is a declaration by a familiar private (or sometimes public) entity and usually professional (but occasionally political) entity in a public place that an event is noteworthy. It suggests that what is published has a call on public attention. Placement on the page or in the broadcast indicates *how* noteworthy; readers and viewers understand the hierarchy of importance this creates. It is a *hierarchy of moral salience*. It is no wonder that the sacred center of the working day on a metropolitan newspaper is the editorial conference to decide what stories will make page one, and where on the page they will go. (pp. 20–21, emphasis added in the latter instance)

Schudson’s depiction reinforces two key ideas that serve as a point of departure for this study: (1) that the hierarchy of news presentation is a moral question, a matter of normative import because of its shaping influence on public attention, as decades of agenda-setting research has demonstrated (McCombs, 2005); and (2) that such prioritizing of news historically has been associated with human deliberations, not machine determinations.

The rise of the digital, mobile, and social media and the emergence of online news during the past 25 years have not, on their own, eliminated traditional modes of news gatekeeping—editors still decide what to publish and where—but they have disrupted longstanding arrangements of prioritization. They have altered control over news hierarchy, resulting in new relationships between people and machines that shape how news is seen, circulated, and interpreted (Carlson, 2018; DeVito, 2017; Lewis & Westlund, 2015a). For example, as gatekeeping becomes more complex and collaborative, ordinary users act as “secondary gatekeepers” in sharing and amplifying news online (Singer, 2014). On social media platforms in particular, users contribute to a fluid process of “networked gatekeeping” by which certain voices and perspectives are prioritized over others (Meraz & Papacharissi, 2013). Machines play an ever-larger role in gatekeeping determinations, as algorithms select hashtags for Twitter trending topics, highlight certain headlines atop Google search results, and determine which items appear prominently in Facebook News Feeds. Digital intermediaries—particularly Google as the dominant search engine and Facebook as the dominant social media platform—control the channels of communication that news publishers once maintained, thereby shaping what kind of news (and advertising) reaches users and on what terms (Bell & Owen, 2017; Nielsen & Ganter, 2017).

Within newsrooms, meanwhile, journalists are actively responding to the new rules set by algorithmic gatekeepers. Search engine optimization (SEO) and social media optimization (SMO) strategies are regularly used by editors at different stages of news production (Dick, 2011; Giomelakis & Veglis, 2016; Poell & van Dijk, 2015; Smyrniaos, 2012). The investment of large and established news organizations in SEO, along with digital audience analytics that indicate to editors how stories are playing on various platforms, influence the dynamics of news distribution—and may contribute to isomorphic tendencies as news organizations mimic one another’s strategies for online presentation (Boczkowski, 2010) and attribute greater editorial value to content that they believe will be favored by external algorithms (Tandoc & Maitra, 2017). The work of modifying headlines, metadata, and other content features for maximum visibility in search results may explain, at least in part, the results presented in this paper. Altogether, how news is prioritized remains a normative question, but such gatekeeping no longer belongs to journalists or humans exclusively.

## 3. Filter bubbles and fragmentation: news consumption in an era of algorithms

Among the many questions raised by machines as news gatekeepers

are two inter-related issues central to debates about news and politics in the internet era: news personalization, on the one hand, and news diversity (or fragmentation), on the other. In the first instance, it is presumed, algorithms may contribute to a narrower range of news content, isolating individuals from a broader set of information that might challenge their beliefs by giving personalized recommendations based on previous clicks, likes, and shares. And, in the second instance, by virtue of their personalized nature, algorithms may contribute to a more fragmented set of news sources—that is, leading users away from traditional news organizations to those niche sources that more closely pertain to their preferences, thereby altering the relative homogeneity or diversity of news media consumption both individually and collectively.

Regarding personalization, a primary concern about algorithms as news gatekeepers is their capacity to tailor information encounters, typically described as leading to echo chambers (Sunstein, 2002, 2018) or “filter bubbles” (Pariser, 2011). This personalization, it is argued, limits what people see and experience, particularly from opposing viewpoints, and ultimately may narrow their knowledge and choices. As algorithms take on a greater array of journalistic tasks, news consumers, it is feared, are in peril of being exposed to a biased, niche, and altogether incomplete picture of the world. The literature broadly touches on the various foci of personalization online, such as search results (Hannak et al., 2013), social media feeds (Bakshy, Messing, & Adamic, 2015; Nikolov, Oliveira, Flammini, & Menczer, 2015), advertising (Sweeney, 2013), and pricing (Bakos, 1998). The prospect of personalization in online news consumption, however, is treated as particularly problematic, with numerous authors concerned about its siloing effect on public conversations (Negroponte, 1995; Shapiro & Richard, 1999; Sunstein, 2002, 2018). In their view, the importance of quality news distribution in a democracy cannot be overemphasized: to make informed decisions, citizens need access to and engagement with a sufficiently diverse body of information about public life.

Emerging research, however, suggests that the threat of filter bubbles has been overblown. Many recent studies on news exposure, exploring multiple platforms, have found the effect of personalization to be smaller than often assumed (Bruns, 2018; Flaxman, Goel, & Rao, 2016; Zuiderveen Borgesius et al., 2016; Fletcher & Nielsen, 2017; Bakshy et al., 2015). While Gentzkow and Shapiro (2011) find that news outlets online have more homogenous audiences than they do offline, they also show that print news has its own strong incentives for polarization, indicating that digital media are far from the only driver of such gaps in news consumption. Similarly, the literature on selective exposure suggests that news consumers actively select biased news by themselves (Iyengar & Hahn, 2009; Stroud, 2011). That is, polarization in news consumption, the tendency to remain within a bubble of personal preferences, might be the result of individual choices at least as much as algorithmic design.

Alongside concerns over siloing effects, there are related worries about the fragmentation and diversity of the news sources to which individuals are exposed. Because digital media platforms lower barriers to entry for new media actors, such as digital-native news startups (Carlson & Usher, 2016), and because they remove geographical and technological limitations to news distribution, the result is an increasingly fragmented news landscape populated by a growing number of sources catering to niche interests. In the United States, for example, nowhere is this concern more evident than in the proliferation of hyperpartisan and fake news sites in the run-up to the 2016 election (Benkler, Faris, Roberts, & Zuckerman, 2017; Lazer et al., 2018). Though users can seek out such sources on their own, algorithms may contribute to such polarization by directing users, in their searches and via their social media feeds, to certain news sources over others—presumably leading people to those sources that more closely align with their personal interests as registered in previous online activity.

As such, the role of algorithms in news distribution has added new to questions that have been debated for several decades regarding the

optimal diversity of sources for news. Many scholars have been critical of homogeneity and concentration in the news industry (Bagdikian, 1987; McChesney, 1999, 2013); others expressed hope that the digitalization of news would create a more diverse and democratized field, empowering small news operations and digital-born outlets that could highlight overlooked points of view (Hindman, 2008; Shirky, 2008). Others consider the digital news landscape (and the cable television landscape before it) excessively fragmented, making it difficult for a shared set of facts to exist in the public square (Levendusky, 2013; Prior, 2007; Williams & Delli Carpini, 2011). Digital media developments, both generally and in the context of algorithms-as-gatekeepers specifically, call up these concerns anew: On the one hand, they represent unprecedented consolidation in news distribution, as an oligopoly of technology companies—Facebook and Google especially—controls the vast majority of news sharing and digital advertising (Bell & Owen, 2017; Nielsen & Ganter, 2017); and, on the other hand, they represent an unprecedented splintering of news distribution across a seemingly limitless number of devices, feeds, apps, and social contexts.

A growing number of studies have focused exclusively on algorithmic news recommendations and their impact on individual news diets. Some of these authors find that reliance on recommendation algorithms actually *increases* rather than decreases diversity in news consumption (Fletcher & Nielsen, *forthcoming*; Schroeder & Kralemann, 2005). Others find evidence of personalization effects and other biases (Diakopoulos, Trielli, Stark, & Mussenden, 2018; Haim, Graefe, & Borsigs, 2018). Collectively, this literature encourages greater awareness about how the use of search engines and aggregators is changing patterns of news exposure (Helberger, Karppinen, & D'Acunto, 2018).

Overall, the concerns raised in the literature indicate the necessity of examining how reliance on news recommendation algorithms reshapes personalization and diversity in news distribution, illustrating the type of judgments and prioritization that algorithms employ and their corresponding impact on news exposure. However, evaluating the meaning of these findings for public life is much more difficult, given that standards for “good” journalism are contested even when they pertain to human conduct alone. Below, we explore how real users experience political news recommendations made by algorithms, and use the empirical findings about news personalization and diversity to discuss broader questions of normative evaluation.

#### 4. Method: a study of Google News recommendations

For the empirical portion of this paper, and to test concerns about algorithmic gatekeeping in news presentation, we examined the extent of personalization in Google News recommendations among a diverse set of participants. Haim, Graefe, and Brosius (2018) also studied personalization in Google News recommendations but did so using user profiles that the researchers created and controlled. By contrast, our study compares real-world searches conducted simultaneously by 168 people across the United States who used their personal accounts. Despite the different methods, our results are very similar to those of Haim et al. (2018). The design proposed here can easily be extended to cover larger samples of thousands across various locations.

Considering the “new geography” of journalism (Reese, 2016)—the shift from clear distinctions between objects of study and levels-of-analysis to an ecology of variegated, interactive networked spaces—we aimed to develop and apply an original method for measuring individual interactions with black-boxed algorithms in the context of digital news consumption. This is a significant methodological problem (Ørmen, 2016). We propose here a simple and relatively affordable method that meets this challenge and allows for gathering data from news consumers using different personalized platforms. Inspired by studies on search results personalization in computer science (Hannak et al., 2013), we asked users to report the results that they were given when conducting identical searches at the same time. We encourage other media scholars to use this design to understand how automated

gatekeepers operate in real-world interactions and how different types of users—varying by geography, demographics, psychographics, and so forth—are served by them. The goal is not to reverse-engineer algorithms, but to estimate their impact on public life by testing how they serve real users who seek political information.

This study measures the degree of personalization that takes place on Google News, a leading source for news discovery that indexes thousands of news sites for its English version and 50,000 news sources overall (Filloux, 2013). In addition to its strength as a news search and recommendation engine, Google News is synchronized with users' activity across other Google applications, among them search, email, web browsing (if they used the Chrome browser), location services, personal drive, and calendar. Therefore, Google News algorithms have the ability to draw on richly detailed profiles of users and match them with news stories from across the web that are likely to be of interest to them. The potential for personalization is considerable. However, Google News exemplifies awareness of the competing requirements of personalization with generalizable measures of newsworthiness. As part of a redesign in May 2018, Google News now offers users the ability to opt out of any personalization in some of the recommendations they receive (Upstill, 2018). Since May 2018, the platform also offers a service of customized five-story news briefings; yet Google News said that the top two stories will be identical for all users, and only the bottom three personalized (Wang, 2018). The experiments reported here were conducted before these features were introduced.

Several weeks before the 2016 Presidential election, two experiments were conducted via Amazon Mechanical Turk (M-Turk), examining the news recommendations that users were suggested when conducting real searches from their personal accounts. M-Turk has been shown to provide more diverse and representative samples relative to traditional approaches of assembling groups for experiments (Mason & Suri, 2012), thus making M-Turk experimental environments potentially more credible as real-world scenarios. M-Turk also enables immediate interaction with a large set of geographically dispersed participants, an affordance that is particularly useful for studying this phenomenon.

The first experiment was conducted October 10, 2016, the day after a highly contentious U.S. presidential debate. The second experiment was conducted on a Friday evening three weeks later (November 2), with no major news events taking place shortly before (i.e., within a 24-h period). In both experiments, participants ( $N = 168$ ) were asked to log in to their personal Google account and search Google News (at [news.google.com](https://news.google.com)) for news on candidates Hillary Clinton and Donald Trump. The requested order of searches was randomized between states. Participants were then asked to copy the first five links they were recommended on each candidate into a form. A small percentage of the results ( $< 5\%$ ) reflected that participants accidentally searched [google.com](https://google.com) instead of [news.google.com](https://news.google.com), and data from these participants was discarded.

To ensure that differences in results could not be attributed to objective shifts in the news cycle, participants had 2 h to complete the task. In both experiments, nearly all results were collected within the first hour. Continuity in the news cycle was ensured by monitoring the news to guarantee that big stories were not breaking during the experiment.

Because our focus was on exposure to political information related to the 2016 election, we sought to ensure that participants were spread among Democratic-leaning, Republican-leaning, and swing states. In both experiments, the sample was dispersed roughly equally from among the following states: New York, California, Texas, North Carolina, Florida, and Ohio. In each state, roughly a third of the participants self-identified to M-Turk as conservative, roughly a third of the participants self-identified as liberal, and roughly a third did not declare political affiliation. (Table 1).

Overall, a dataset of 1653 links that were recommended to searchers was collected, representing 141 articles from 51 news organizations.

**Table 1**  
Characteristics of the sample ( $N = 168$ ).

|                       |                                                                                                                                                        |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Political Inclination | Conservative: 27%<br>Liberal: 41%<br>Not identified: 32%                                                                                               |
| State                 | New York: 13%<br>California: 21%<br>Texas: 18%<br>Florida: 18%<br>Ohio: 18%<br>North Carolina: 13%                                                     |
| Age                   | 18-35: 55%<br>36-55: 36%<br>56-75: 9%                                                                                                                  |
| Gender                | Female: 39%<br>Male: 61%                                                                                                                               |
| Education Level       | No credentials: 2%<br>High school diploma or equivalent: 29%<br>Some postsecondary education, no degree: 61%<br>BA, Associate degree or equivalent: 9% |

Note: Percentages may not add up to 100% due to rounding.

Our hypothesis was that a significant degree of personalization would be evident in the results, either on the basis of geolocation or (more likely) in connection with searchers' political inclinations that presumably would have been manifest in their previous activities using Google services.

## 5. Results of the experiments

Despite the great potential for personalization in Google News search recommendations, results were quite similar overall. Searchers of various political leanings, across the country, were offered a largely unified body of news from a small number of national publications. Neither ideological bias nor geographic bias were evident in the search results. And, when controlling for other individual-level variables, such as gender, age and location, no significant differences were evident in the news recommended to different participants.

A certain degree of personalization was identified in a minuscule subset of the results. In the first experiment, only conservatives were recommended stories from Fox News, and only liberals were recommended stories from *The Daily Mail*, *New York Daily News*, Salon and The Daily Beast. In the second experiment, only conservatives were recommended stories from *The New York Post*, Business Insider, and Jezebel, and only liberals were recommended stories from *The Washington Times*, *People*, U.S. News & World Report and *US Weekly*. These results were counterintuitive and in fact contradict the typical concern over filter bubbles: While *The New York Post* is known to lean right, Jezebel is anything but conservative in its politics, and likewise *The Washington Times* is anything but targeting liberals. Either way, these differences were marginal, representing less than 0.01% of the data collected.

The more significant finding is that publishers representing 99.9% of the links were recommended to both conservatives and liberals, and the most recommended five articles were nearly identical for both groups. While slight variation was evident in the long tail of the recommendations, they were almost uniform overall. Out of 47 news publishers that Google News recommended to the participants, 41 were recommended to both conservatives and liberals. The other 6 outlets together accounted for less than 0.01% of the recommendations.

These findings partially align with those of Fletcher and Nielsen (forthcoming), who found that usage of search engines for news discovery does not lead to echo chambers, but instead increases diversity in individual news diets, as it tends to introduce searchers to sources they would not have encountered otherwise. Yet the results reported here differ from Fletcher and Nielsen (forthcoming) in one significant respect. While they found that searching for news on search engines generates pluralism, we find that the body of news stories suggested to



**Table 2**  
News outlets that ranked among the Top 5 most popular stories across Google News searches.

| Outlet                  | Total number of recommendations | % of all recommendations | Headquarters (metro area) | Original medium |
|-------------------------|---------------------------------|--------------------------|---------------------------|-----------------|
| The New York Times      | 286                             | 17%                      | NYC                       | Newspaper       |
| CNN                     | 154                             | 9%                       | Atlanta                   | TV              |
| Politico                | 148                             | 9%                       | Washington, DC            | Digital         |
| The Washington Post     | 142                             | 9%                       | Washington, DC            | Newspaper       |
| HuffPost                | 88                              | 5%                       | NYC                       | Digital         |
| Business Insider        | 73                              | 4%                       | NYC                       | Digital         |
| The Wall Street Journal | 70                              | 4%                       | NYC                       | Newspaper       |
| TIME                    | 69                              | 4%                       | NYC                       | Magazine        |
| Fortune                 | 61                              | 4%                       | NYC                       | Magazine        |
| The Chicago Tribune     | 61                              | 4%                       | Chicago                   | Newspaper       |
| ABC                     | 49                              | 3%                       | NYC                       | TV              |
| CNBC                    | 44                              | 3%                       | NYC                       | TV              |
| The Los Angeles Times   | 42                              | 3%                       | Los Angeles               | Newspaper       |
| USA Today               | 17                              | 1%                       | Washington, DC            | Newspaper       |

participants reflected a significant degree of homogeneity and concentration in automated news distribution. That is, even if people were recommended news from sources outside their partisan preference, they were nevertheless recommended almost entirely the same set of news sources.

In total, the four searches that we examined—one each for Clinton and Trump, repeated twice—yielded recommendations ranging from 16 to 32 news outlets, with an average of 23.25. But despite the large number of news sources suggested to searchers, the vast majority of recommendations clustered around a small number of news sources. In each experiment, 50%–85% of all news recommendations led to the same five news sites. On average, the top five outlets accounted for 69% of all links. In other words, when a diverse set of people across the country searched for news about the presidential candidates, they were regularly referred to a set of five news sources alone (whose identity changed in each search), despite the platform's algorithmic capability of constructing a much more diverse and/or tailored news experience. (see Figs. 1–4).

Across the four searches, fourteen news organizations ranked in the top five: CNN, Politico, *Fortune*, *The Chicago Tribune*, Business Insider, *The New York Times*, *The Wall Street Journal*, *The Washington Post*, CNBC, ABC, *Time*, *The Los Angeles Times*, HuffPost, and *USA Today*. Together, these outlets made up 79% of the total number of news recommendations suggested to searchers. (Table 2).

The number of recommendations was unevenly distributed among these organizations as well, with the five most prominent outlets—*The New York Times*, CNN, Politico, *The Washington Post* and HuffPost—making up half (49%) of the 1653 total recommended links. In other words, while Google News indexes many thousands of English-language news sources, every time our participants searched for news, they had a 49% chance of being directed to one of these five outlets.

There is some degree of diversity among the fourteen news organizations that occupied the top spots in the two experiments. Some are legacy print media, some are digital-native, and others are television networks and magazines. Some are general interest outlets, and others are focused on the political world. But almost all of them are large national organizations, and 79% of them are based in only two metro areas: New York City and Washington, D.C. (with the exceptions headquartered in the Atlanta, Chicago, and Los Angeles areas). These findings indicate that more than dissecting the public conversation into fragmented silos, digital distribution of news might narrow the conversation around a small set of national outlets.

Moreover, the fact that 11 of the top 14 outlets were newspapers and television networks and only 3 were digital-native sites indicates that the news agenda suggested by Google News largely maintains, rather than disrupts, the reach and influence of legacy news organizations that came to prominence before the digital age. Our findings suggest that Google News rankings highlight longstanding national

newspapers and television networks over local and younger news organizations. In effect, the news experience that Google News constructed for users in these experiments is highly concentrated, empowering a handful of prominent outlets and marginalizing others. This information landscape resembles a “high-modern,” late-twentieth century mass media news diet (cf. Hallin, 1992) much more than the popular image of digital news as a fragmented, diverse, and personalized terrain—particularly when digital news presentation is being made by algorithms acting as news gatekeepers.

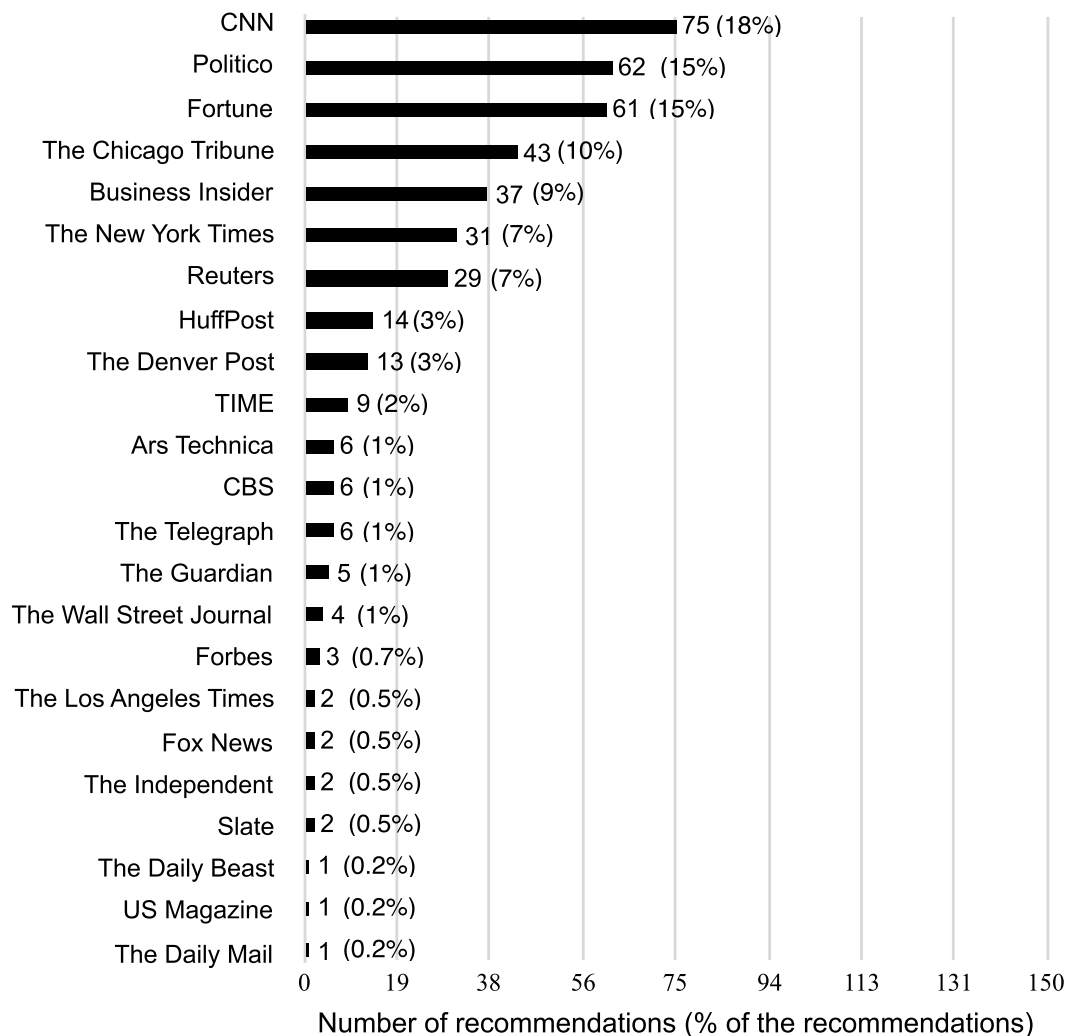
## 6. Analysis of the results

Overall, our findings indicate that despite the ability of algorithms to provide much more personalized headlines than human editors, they might actually produce, at least in some cases, highly centralized and unified news diets across diverse sets of users. These data were collected when the public conversation about the spread of fake news (and corresponding concerns about algorithmic filter bubbles) was still nascent. But it seems that, even as of fall 2016, various social, economic, and political concerns—among others invisible to us as researchers—may have functioned as a check on news algorithms, leading them to not fully manifest their technological capability for personalization. However, we examined but one news recommendation engine. The degree of personalization may differ among platforms, and further research is needed to map these potential differences.

These findings correspond with those of Haim et al. (2018), who also examined personalization on Google News by constructing controlled accounts. They also found minuscule differences between the news stories suggested to different “profiles,” and found evidence for algorithmic prioritization of certain news outlets. Some of the potential explanations they suggest, such as Google potentially punishing outlets that use paywalls, do not seem relevant for our findings. Others, such as potential correlations between placement on Google News and organizational investment in search engine optimization (SEO), might well apply here. Our findings also correspond with Brown's (2018) analysis of recommendations on Apple News, which, although curated by editors, similarly favor and reward “a small group of big players.”

These findings raise several questions. While Google emphasizes that Google News is computer-generated and algorithmic (Google, 2013, 2016), without human intervention, are algorithmic logics the sole driver behind these patterns of prioritization? Do Google News search results really offer “a wide variety of perspectives on any given story,” as Google (2013) assures? Do they reflect the range of the public conversation on central topics? Might search results be programmed to avoid too heavily mirroring individual users' preferences and interests, so as to escape the appearance of overt bias?

Just as important, how should these findings be evaluated against a normative standard? Are these patterns of homogeneity “good,” or



**Fig. 1. News Recommendations in Google News searches for “Hillary Clinton” on October 10, 2016.** Note: The top 5 outlets accounted for 67% of all news stories recommended. Searches were conducted by 84 participants in New York, California, Texas, Florida, Ohio, and North Carolina between 2–4 p.m. EDT. Of the participants, 41% identified as liberal, 26% as conservative, and the remaining 33% did not identify a partisan leaning. Percentages may not add up to 100% due to rounding.

not—and why? How might the presentational patterns of machines as gatekeepers be evaluated for what they contribute (or not) to the quantity and quality of information about public affairs? Addressing this question, as we explore in the section below, is surprisingly difficult, largely because the normative standard for good journalism is itself quite contested.

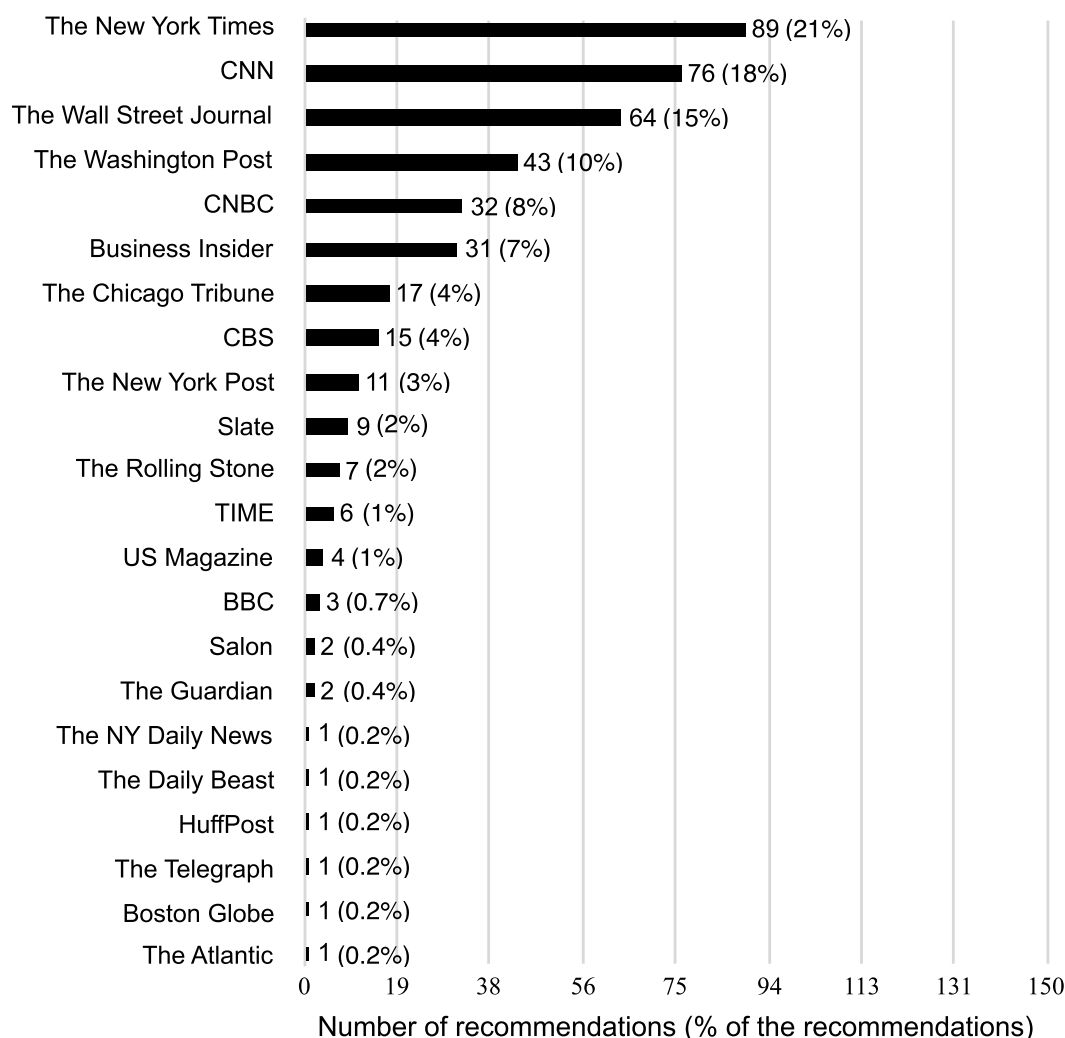
## 7. Discussion: toward a normative evaluation of machines as news gatekeepers

In illustrating what machines may be doing as news gatekeepers, our experiments gesture to the need for exploring larger questions about why they are operating as such—and, even more, what news producers and consumers should want from algorithmic editors. In popular imagination, there is persistent, if perhaps misplaced, hope that machines may perform as well as, and perhaps even better than, humans in certain tasks, such as the objective presentation of news. But what exactly does it mean to perform “well” as a journalist, human or otherwise? This is where the discussion becomes complex, making the evaluation of empirical findings on the conduct of algorithms in news as challenging as it is crucial.

While the rise of machines in journalism has ethical implications (Lewis & Westlund, 2015b), it can be difficult to qualify them. The

conduct of humans is often the standard against which machines are compared. However, journalism is one field in which normative standards for humans already are in a seemingly permanent state of flux, with continuous negotiations around the boundaries of what counts as news and who counts as a journalist (Carlson & Lewis, 2015). When evaluating driverless cars, the parameters are comparatively simple: automated cars should reach destinations safely and efficiently. Despite concerns around privacy and data security, assessing the quality of machine behavior in the field of driving adheres to relatively straightforward standards. Likewise, when evaluating the conduct of machine caretakers, it is not very difficult to assess whether they are fulfilling their aim of improving patient health. In journalism, however, qualitative assessments are generally more contested.

Questions of personalization and fragmentation in news are exemplary cases of this challenge. Discussions of these problems are largely driven by, and reflective of, competing normative visions. The concerns frequently expressed over the fragmentation of online news consumption, for example, reflect a hypothesis that a healthy, public-serving news market is characterized by a sufficiently unified news agenda, one that makes a standardized body of important information accessible throughout society (Gitlin, 2002; Prior, 2007; Sunstein, 2002). However, others argue that democratic public life benefits from a richness of voices and viewpoints, and that the journalism landscape



**Fig. 2. News Recommendations in Google News searches for “Donald Trump” on October 10, 2016.** Note: The top 5 outlets accounted for 73% of all news stories recommended. Searches were conducted by 84 participants in New York, California, Texas, Florida, Ohio, and North Carolina between 2–4 p.m. EDT. Of the participants, 41% identified as liberal, 26% as conservative, and the remaining 33% did not identify a partisan leaning. Percentages may not add up to 100% due to rounding.

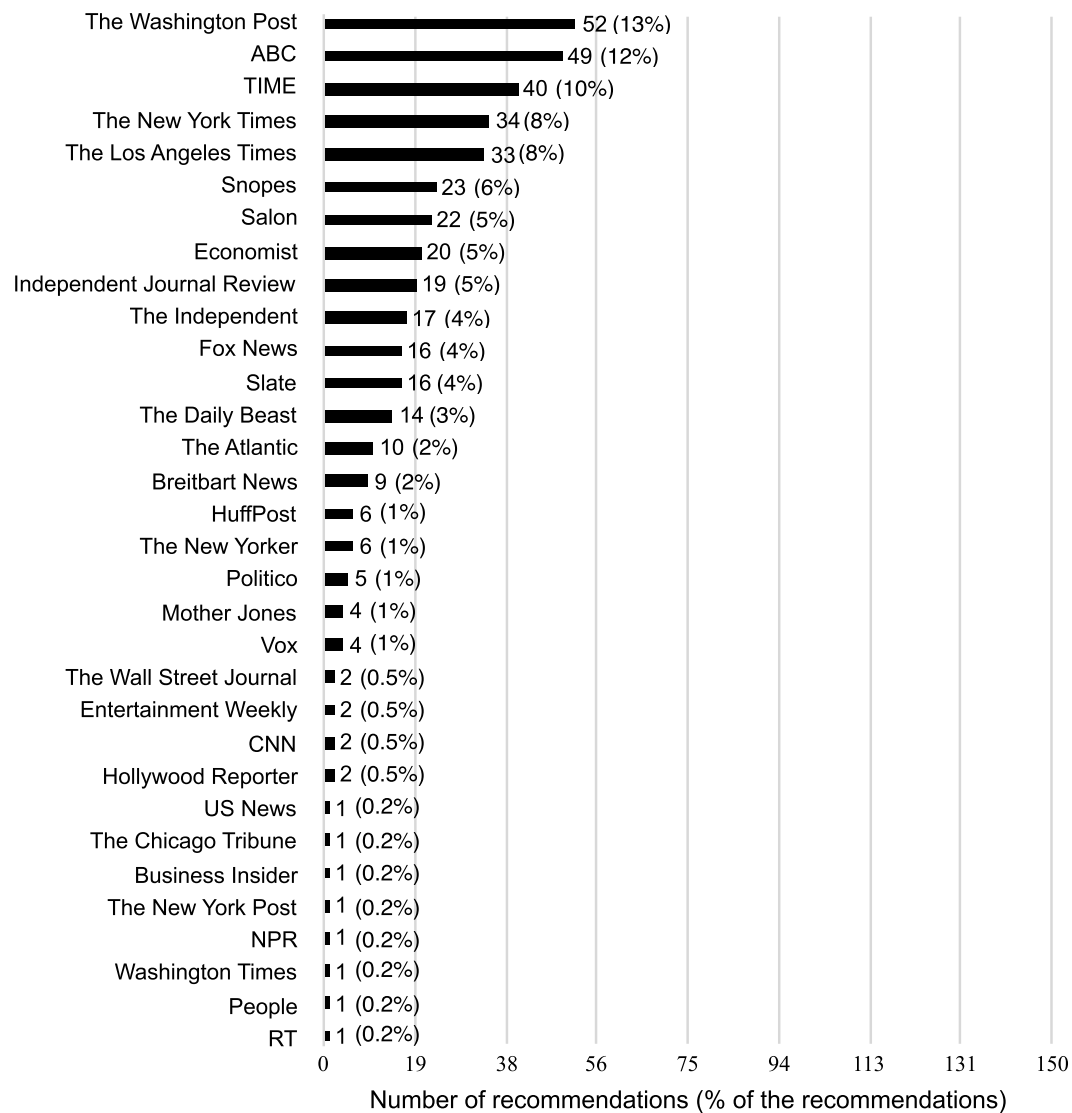
should reflect social, organizational and epistemic diversity (Anderson, 2016; Benson, 2013; McQuail, 1992; Voakes, Kapfer, Kurpius, & Shano-yeon Chern, 1996). Should one hope, then, that algorithmic news distribution generates homogeneity, or at least a degree of similarity, in news consumption and agenda—or is the public best served if algorithms highlight varied information from a large number of sources that offer a potpourri of perspectives? Empirical research cannot truly answer these questions, as there is no agreed-upon standard for an “optimal” degree of personalization or pluralism in news.

Despite this normative plurality, even in journalism it is generally easy to tell when machines are failing egregiously. If every news story recommended by a search engine were false, or if a search engine referred readers to one news organization alone, it would be clear that the algorithm is falling short as a news provider acting in the public interest. But, in most cases, public-facing algorithms of this sort do not function catastrophically or perfectly, but somewhere in between. Precisely where they fall on such a spectrum remains open for debate. What standards should be used to assess their performance? Which visions of the public and of public life should such algorithms embody and encourage (cf. Anderson, 2011)?

Such questions may be frustratingly hard to grasp without a more thorough understanding of algorithms' design and deployment. While such matters are usually beyond researchers' reach, continually

wrestling with these lines of inquiry is an imperative element of critical algorithm studies (Ziewitz, 2016). Many scholars have commented on the importance of going beyond description to deal with normative implications when studying technological actants (Carlson, 2017; Gillespie, 2010; 2014; Kraemer, Van Overveld, & Peterson, 2011; Nissenbaum, 2001). Because journalism is a deeply normative activity, descriptions alone are not sufficient for discussing the changing roles of news in society (Benson, 2017; Blumler & Cushion, 2014). The conduct of machines should be examined and described with rigor, but then evaluated against a normative backdrop, even if this backdrop remains contested. This is particularly true at a time when the emerging role of machines as communicators challenges longstanding assumptions about what is distinctly “human” about journalism as both a functional form of communication and a normative exercise in public meaning-making.

Journalism is not the only field characterized by this kind of normative plurality. Other fields face similar challenges in evaluating the conduct of machines: they likewise might recognize when machines fail badly, and yet struggle to define what the normal standard should be. Consider, for example, the use of machines in law enforcement and criminal justice, in which there is intense debate about the proper role of algorithms in guiding police behavior, targeting prospective criminals, and determining prison sentences (e.g., Brayne, 2017; O'Neil,



**Fig. 3. News Recommendations in Google News searches for “Hillary Clinton” on November 2, 2016.** Note: The top 5 outlets accounted for 50% of all news stories recommended. Searches were conducted by 84 participants in New York, California, Texas, Florida, Ohio, and North Carolina between 5:30–7:30 p.m. EST. Of the participants, 40% identified as liberal, 27% as conservative, and the remaining 33% did not identify a partisan leaning. Percentages may not add up to 100% due to rounding.

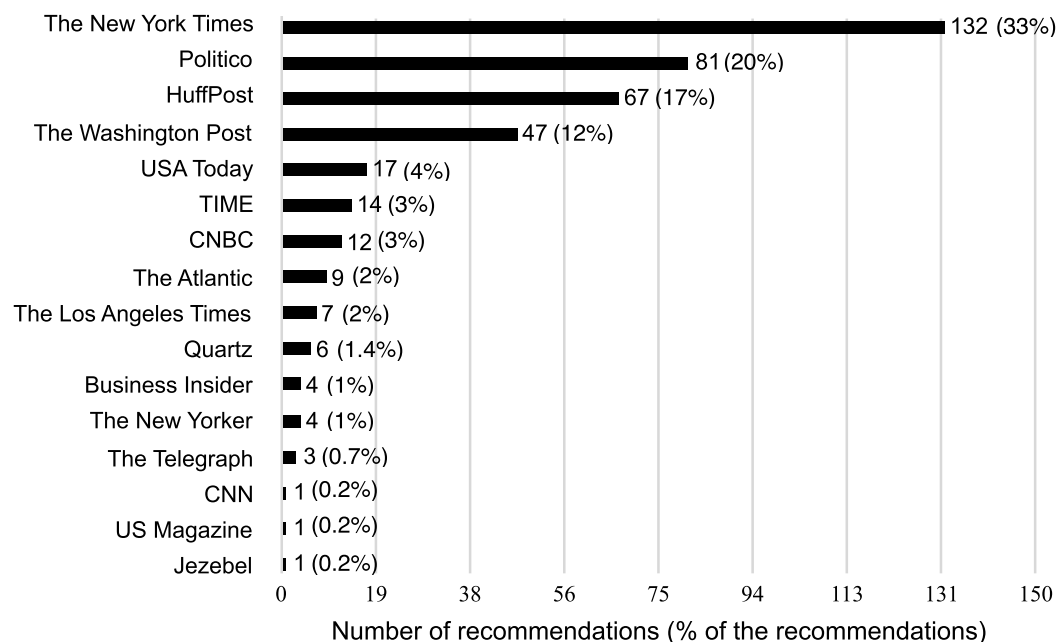
2016). It is possible, and necessary, to measure whether judgments made by machines in these instances are generally similar to those made by humans. But are the judgments made by humans optimal to begin with? In criminal justice as in journalism, it can be relatively simple to call out outrageously unjust or improper decisions, but ordinarily the conduct of both humans and machines is interpretative and subjective (Carlson, 2017). As a senior judge said to Christin (2017, p. 9), “You can take the same case, with the same defendant, the same criminal record, the same judge, the same attorney, the same prosecutor, and get two different decisions in different courts ... Is that justice? I think it is.”

Politics is an equally contested field, in which normative pluralism is an essential component. The role of machines in democratic elections, for example, is at the heart of a much older debate than the one about algorithms: There remains no consensus on the proper roles for machines in voting, and whether machines are even necessary at all, with many countries and localities continuing to use paper balloting (Birch, Cockshott, & K Renaud, 2014). Evaluating the conduct of machine actors in voting requires normative judgment alongside data, as different political systems prioritize different values in the electoral process.

In business as well, firms, regulators, and other stakeholders face challenges around the management of machines in different processes, from privacy practices to artificial intelligence training and conduct. These dilemmas are difficult to answer definitively when technological developments render existing standards inadequate in an already variegated field (Ekbia et al., 2015). Similarly, computer-generated art is a creative field in which the evaluation of human production is, as a rule, subjective and flexible, and so the evaluation of machine-produced elements creates new problems (Gunkel, 2017).

Altogether, these are rather different types of human activity, practiced, organized, and regulated in different ways. But they share, perhaps, a “sense of dissonance” (Stark, 2011), or an appreciation for the generative potential of competing notions of value. Contrary to convention wisdom that differences inherently lead to conflict and stagnation, Stark (2011) argues that “it is through *unshared* typifications, through uncommon attributions, through divergent or misaligned understandings that problematic situations can give way to positive reconstructions” (p. 192, emphasis original). As such, these and other occupational fields may have rival interpretations, multiple perspectives, and competing forms of judgment; these may lead to productive





**Fig. 4. News Recommendations in Google News searches for “Donald Trump” on November 2, 2016.** Note: The top 5 outlets accounted for 85% of all news stories recommended. Searches were conducted by 84 participants in New York, California, Texas, Florida, Ohio, and North Carolina between 5:30–7:30 p.m. EST. Of the participants, 40% identified as liberal, 27% as conservative, and the remaining 33% did not identify a partisan leaning. Percentages may not add up to 100% due to rounding.

debates within and among these domains, but at the same time complicate our ability to evaluate machine behavior from a normative standpoint.

In conclusion, this article has sought to do two things: report on a study that implements an innovative and effective method for measuring algorithmic personalization in online news recommendations, and reflect more broadly on the challenges of determining normative standards for establishing what “good” machines do when they stand in, as they increasingly do, as key gatekeepers of the news media environment. The central role of machines in news production, distribution, and consumption makes normative judgments of their conduct an essential topic of research. In news markets, as in many other domains of human activity, machines—and the cultural, political, and economic interests behind them—are reshaping the landscape faster than scholars and practitioners can assess their social impact, particularly on matters of quality, diversity, and plurality. Because these are formative times for the integration of machines into news production and circulation, reshaping how news is organized, prioritized, and made visible, empirical and normative investigations are especially needed. At the same time, this article suggests that because there is no one agreed-upon standard for humans as news gatekeepers, assessing the performance of machines in that role is doubly complicated.

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